

CHENNAI FLOODS THROUGHOUT THE YEARS (2015-2023)

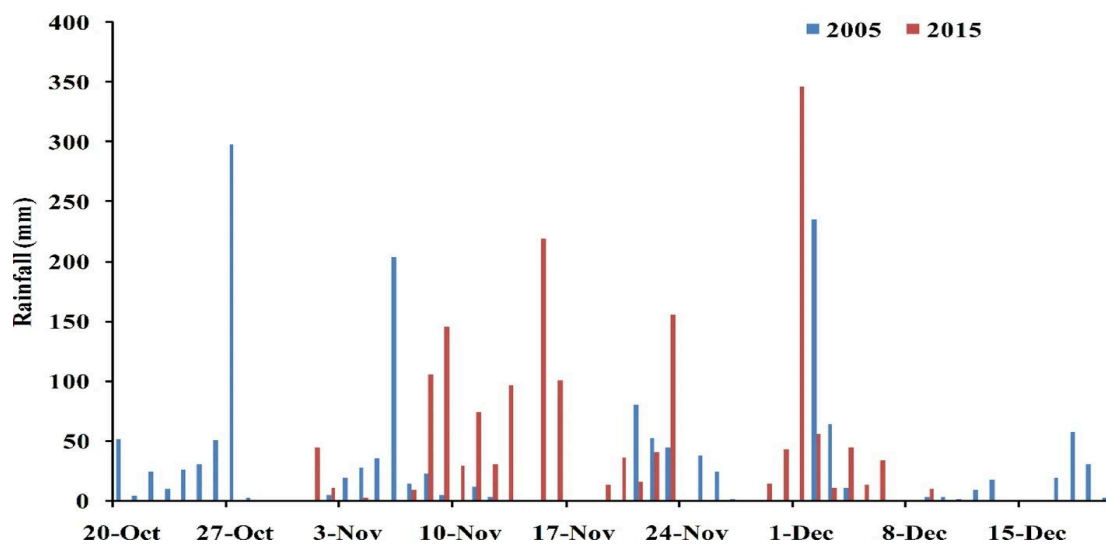
-BY A LEO PRINCE

Chennai ,known famously as the detroit of asia is also known for its destructive floods which seem to halt everyday life every monsoon.

In this article we take on our detective glasses and try to find its patterns .Perhaps we could find new ways to prevent or minimise the damages through mindful decisions.

2015 floods

It was one of the biggest floods faced by chennai caused due to relentless monsoon rains ,exacerbated by poor city planning and the sudden release of Chemberabakkam reservoir leading to overflow of the Adyar river and the Coovum.



```
mysql> select count(locality) from floods where year=2015 and flood in ('heavy','moderate');
+-----+
| count(locality) |
+-----+
|                20 |
+-----+
```

Over 20 localities were moderately or heavily flooded.

2016 floods

Caused by cyclone Vardah

```
mysql> select count(locality) from floods where year=2016 and flood in ('heavy','moderate');
+-----+
| count(locality) |
+-----+
|                5 |
+-----+
```

Over 5 localities were moderately or heavily flooded.

2017 floods

Caused due to low pressure area during monsoon season

```
mysql> select count(locality) from floods where year=2017 and flood in ('moderate','heavy');
+-----+
| count(locality) |
+-----+
|               16 |
+-----+
```

Over 16 localities were moderately or heavily flooded.

2020 floods

Caused by cyclone Niwar

```
mysql> select count(locality) from floods where year=2020 and flood in ('heavy','moderate');
+-----+
| count(locality) |
+-----+
|                5 |
+-----+
```

Over 5 localities were moderately or heavily flooded.

2021 floods

Caused by a cyclonic circulation in the Bay of Bengal.

```
mysql> select count(locality) from floods where year=2021 and flood in ('heavy','moderate');
+-----+
| count(locality) |
+-----+
|                17 |
+-----+
```

Over 17 localities were moderately or heavily flooded.

2023 floods

Caused by cyclone Michuang

```
mysql> select count(locality) from floods where year=2023 and flood in ('heavy','moderate');
+-----+
| count(locality) |
+-----+
|                9 |
+-----+
```

Over 9 localities were moderately or heavily flooded.

OVERVIEW:

```
mysql> select sum(area_in_km2),zone from floods group by zone order by sum(area_in_km2) desc;
+-----+-----+
| sum(area_in_km2) | zone |
+-----+-----+
| 334.74 | ADYAR RIVER BANK |
| 174.52 | NORTH & WEST |
| 131.02 | IT CORRIDOR & SOUTH |
| 59.25 | CENTRAL & LOW LYING |
| 5.86 | NORTH & LOW LYING |
+-----+-----+
```

The above table shows the sum of all the areas affected during all the floods grouped with their zone within chennai.

We can see that areas around Adyar river bank are the most affected .This points that excessive incessant rainfall makes the river banks prone to flooding due to overflow.

```
mysql> select avg(area_in_km2),zone from floods group by zone order by avg(area_in_km2) desc;
```

avg(area_in_km2)	zone
16.737000	ADYAR RIVER BANK
7.271667	NORTH & WEST
5.955455	IT CORRIDOR & SOUTH
4.232143	CENTRAL & LOW LYING
1.953333	NORTH & LOW LYING

The table shows the average area of the localities in different zones of the city vary considerably .This should be taken into account when considering the total area affected comparison.

```
mysql> select sum(area_in_km2),flood from floods group by flood;
```

sum(area_in_km2)	flood
593.96	HEAVY
27.02	NO
10.88	LOW
9.00	NULL
40.67	MODERATE
23.86	LIGHT

The table shows the total areas affected by different calibre of flooding over the years.

The solution

The state government invests heavily in SWDs(Storm Water Drainage) systems across the city.

The following is the status of the localities

locality	drainage
MUDICHUR	N
TAMBARAM	N
AKAPUTHUR	N
WEST MAMBALAM	Y
KOTTURPURAM	N
JAFFERKHANPET	N
VELACHERY	O
T.NAGAR	Y
SAIDAPET	Y
MYLAPORE	N
ROYAPETTAH	N
PERUMBAKKAM	N
SHOLINGNALLUR	N
PALLIKARNAI	N
ALONG OMR	NULL
KORATTUR	Y
AMBATTUR	Y
MADHAVARAM	Y
VYASARPADI	Y
RED HILLS	N
TONDIARPET	Y
PERAMBUR	Y
KODUNGAIYUR	N
KILPAUK	N
ALANDUR	Y
EEKADUTHANGAL	N
NUNGAMBAKAM	Y
GUINDY	N
TARAMANI	N
VEPERY	Y

KONDUNAIGAYUR	N
VYARSARPADI	Y
OTTERI	Y
NUNGAMBAKKAM	Y
TIRUSULAM	N
ADYAR	Y
MADIPAKKAM	Y
ADAMBAKKAM	Y
AYANAVARAM	Y
VILLIVAKAM	Y
EGMORE	Y
BABA NAGAR	Y
GKM COLONY	Y
JAWAHAR NAGAR	O
PAPER MILLS ROAD	Y
KUNDRATHUR	N
THIRUMUDIVAKKAM	N
THIRUNEERMALAI	N
SIRUKALATHUR	N
KOLATHUR	Y
CHETTIMEDU	N
PERUNGAVUR	N
VICHOOR	N
INJAMBAKKAM	O
ALWARPET	Y
THORAIPAKKAM	O

legend:

Y=yes

N=no

O=partially

While some of the areas have received coverage others have been neglected.

```
mysql> select sum(area_in_km2) from floods where drainage='y';
+-----+
| sum(area_in_km2) |
+-----+
|          204.70 |
+-----+
```

The total area affected by floods covered by SWD(2015-2023).

Note:The SWD data is from 2023

So area under SWD coverage affected by floods has decreased considerably in 2023

```
mysql> select sum(area_in_km2) from floods where drainage='y'and year=2023;
+-----+
| sum(area_in_km2) |
+-----+
|          17.95 |
+-----+
```

```
+-----+
| sum(area_in_km2) |
+-----+
|          81.78 |
+-----+
```

in 2015

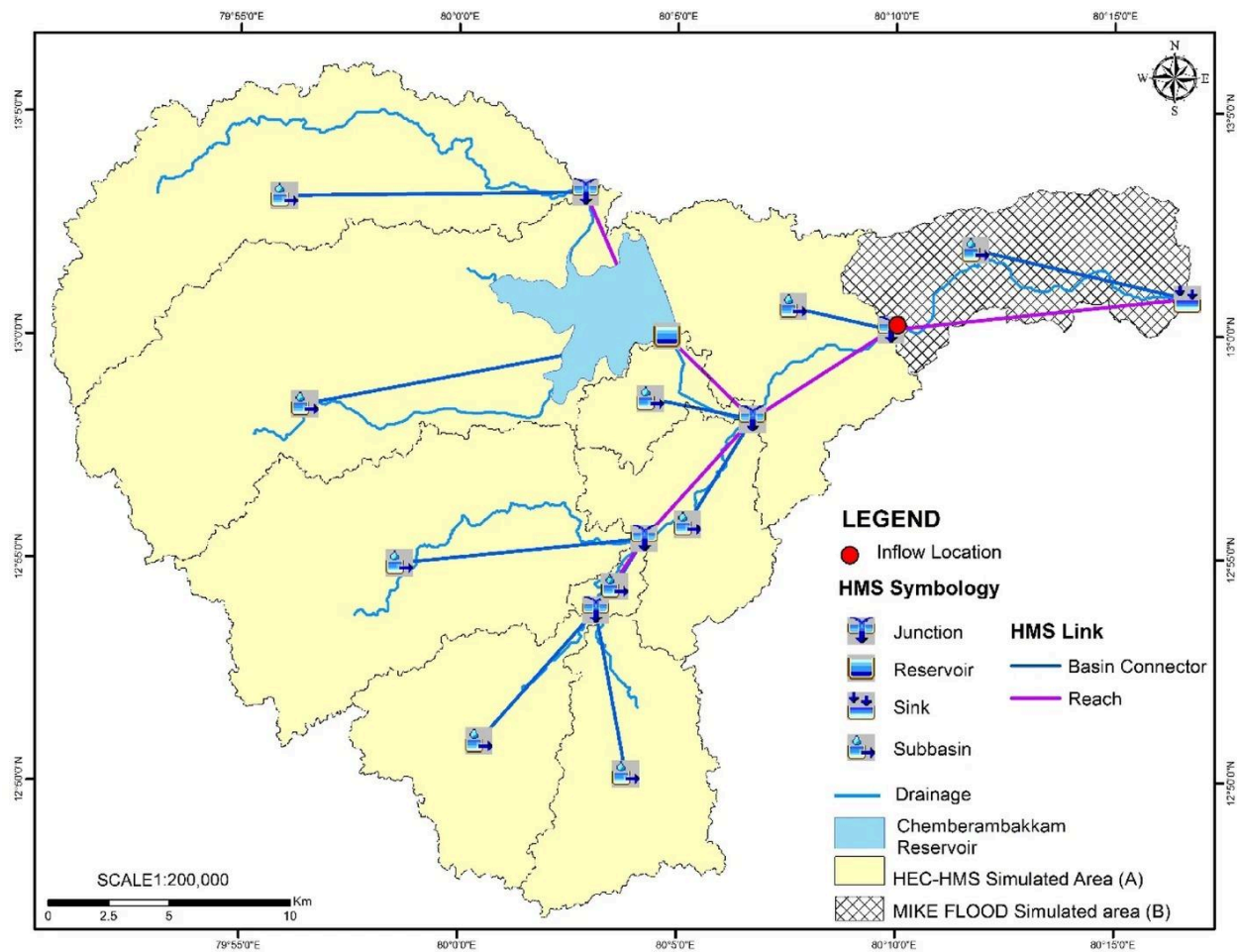
Where there is no drainage(SWD) in 2023:

```
mysql> select sum(area_in_km2) from floods where drainage='n'and year=2023;
+-----+
| sum(area_in_km2) |
+-----+
|          36.85 |
+-----+
```

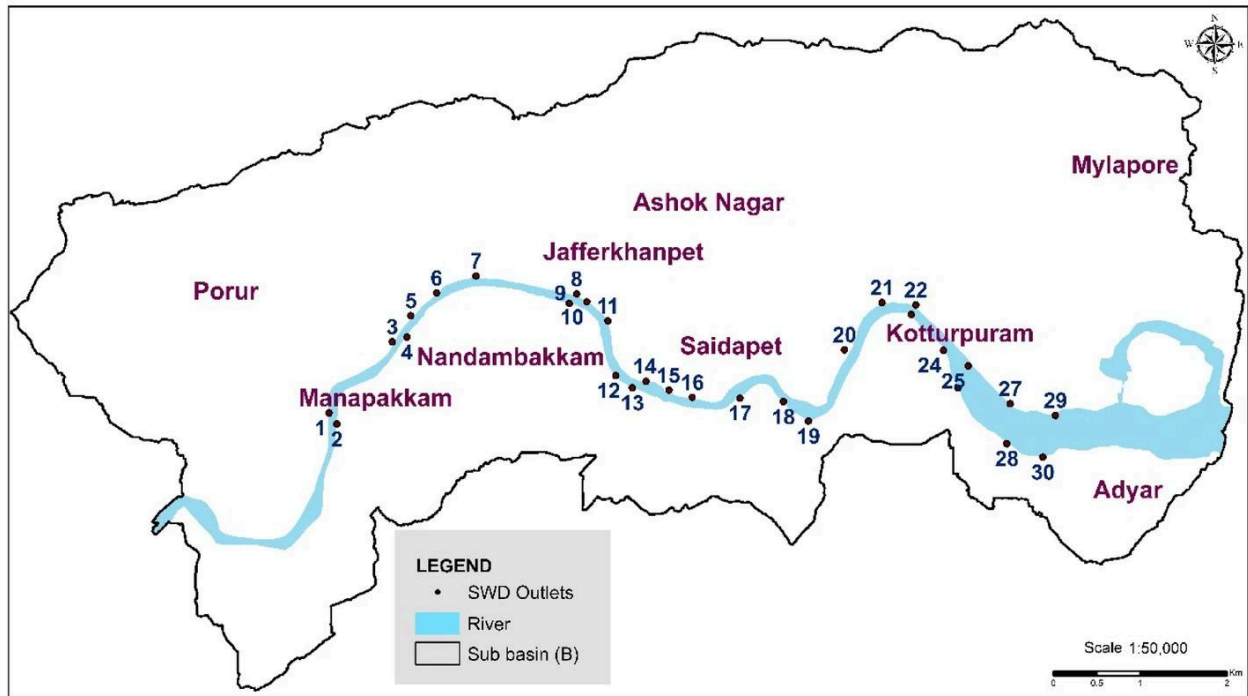
Lets see in it in 2015 just in case

```
mysql> select sum(area_in_km2) from floods where drainage='n'and year=2015;
+-----+
| sum(area_in_km2) |
+-----+
|          146.84  |
+-----+
```

Here,



(1)



(2)

The above figures show a pictographic representation of Storm Water Drainage outlet(1) and the basins in which the water gets stored in or let go(1).

Raj Bhagat P, a civil engineer and geo-analytics expert said better stormwater drainage systems in the city would not have been able to prevent the flooding.

"This solution would have helped a lot in moderate and heavy rainfall, but not in very heavy and extremely heavy rains," he said.

So SWDs are sure helpful and help in decreasing flooding risk but only upto a point.

But proper maintenance and cleaning is crucial for them to work at full capacity. A lack of which causes severe waterlogging in main roads such as Taramani Link road and subways such as Gugu Reddy Subway.

The zones with the most affected localities are given below.

```
mysql> select count(locality), zone from floods where flood in ('heavy','moderate', 'light') group by zone order by count(locality) desc;
```

count(locality)	zone
23	IT CORRIDOR & SOUTH
21	NORTH & WEST
17	ADYAR RIVER BANK
13	CENTRAL & LOW LYING
3	NORTH & LOW LYING

Below are are number of most affected areas(localities) in each zone.

```
mysql> select COUNT(LOCALITY),ZONE FROM FLOODS WHERE MOST_AFFECTED='Y' GROUP BY ZONE;
```

COUNT(LOCALITY)	ZONE
10	ADYAR RIVER BANK
18	IT CORRIDOR & SOUTH
8	CENTRAL & LOW LYING
20	NORTH & WEST
2	NORTH & LOW LYING

Even though IT corridor and South has more localities affected ,North and West has more which are severely affected.

MY DATABASE

sno	locality	area_in_km2	YEAR	flood	most_affected	DRAINAGE	RAIN_IN_CM	zone
1	MUDICHUR	3.43	2015	HEAVY	Y	N	40	ADYAR RIVER B
2	TAMBARAM	87.64	2015	HEAVY	Y	N	49	ADYAR RIVER B
3	AKAPUTHUR	3.84	2015	HEAVY	Y	N	49	IT CORRIDOR &
4	WEST MAMBALAM	2.00	2015	HEAVY	Y	Y	~50	ADYAR RIVER B
5	KOTTURPURAM	2.74	2015	HEAVY	Y	N	~50	ADYAR RIVER B
6	JAFFERKHANPET	3.02	2015	HEAVY	Y	N	35	ADYAR RIVER B
7	VELACHERY	6.82	2015	HEAVY	Y	O	40	CENTRAL & LOW
8	T.NAGAR	4.57	2015	HEAVY	Y	Y	33+	CENTRAL & LOW
9	SAIDAPET	2.87	2015	HEAVY	Y	Y	47 CHEMBARABAKAM RESERVOIR FLOW	ADYAR RIVER B
10	MYLAPORE	3.91	2015	HEAVY	Y	N	25-35	CENTRAL & LOW
11	ROYAPETTAH	1.48	2015	HEAVY	Y	N	~29.4	CENTRAL & LOW
12	PERUMBAKKAM	6.00	2015	HEAVY	Y	N	38-49	IT CORRIDOR &
13	SHOLINGNALLUR	11.44	2015	HEAVY	Y	N	NULL	IT CORRIDOR &
14	PALLIKARNAI	13.42	2015	HEAVY	Y	N	40+	IT CORRIDOR &
15	ALONG OMR	NULL	2015	HEAVY	Y	NULL	NULL	IT CORRIDOR &
16	KORATTUR	7.85	2015	HEAVY	Y	Y	33.6	NORTH & WEST
17	AMBATTUR	45.00	2015	HEAVY	Y	Y	22.3-27.2	NORTH & WEST
18	MADHAVARAM	14.21	2015	HEAVY	Y	Y	35-49	NORTH & WEST
19	VYASARPADI	5.28	2015	HEAVY	Y	Y	28-34	IT CORRIDOR &
20	RED HILLS	9.92	2015	HEAVY	N	N	34	NORTH & WEST
21	SAIDAPET	2.87	2016	NO	N	Y	~16-17	ADYAR RIVER B
22	AMBATTUR	4.50	2016	LOW	N	Y	~16-17	NORTH & WEST
23	TONDIARPET	4.38	2016	LOW	Y	Y	~16-17	NORTH & WEST
24	PERAMBUR	6.33	2016	HEAVY	Y	Y	14	NORTH & WEST
25	KODUNGAIYUR	9.00	2016	NULL	N	N	~16-17	NORTH & WEST
26	KILPAUK	3.97	2016	NO	N	N	~16-17	CENTRAL & LOW
27	ALANDUR	2.00	2016	LOW	N	Y	~16-17	ADYAR RIVER B

28	EEKADUTHANGAL		2.38	2016	NO	N	N	~16-17	IT CORRIDOR &
SOUTH									
29	MUDICHUR		3.43	2016	MODERATE	Y	N	~16-17	ADYAR RIVER B
ANK									
30	PERUMBAKKAM		6.00	2016	HEAVY	Y	N	~16-17	IT CORRIDOR &
SOUTH									
31	RED HILLS		9.92	2016	NO	N	N	9.3	NORTH & WEST
32	NUNGAMBAKAM		3.38	2016	MODERATE	N	Y	21	IT CORRIDOR &
SOUTH									
33	GUINDY		7.88	2016	NO	N	N	16	ADYAR RIVER B
ANK									
34	TARAMANI		3.72	2016	MODERATE	N	N	13	IT CORRIDOR &
SOUTH									
35	VEPERY		1.05	2017	HEAVY	Y	Y	8-20	NORTH & WEST
36	PERAMBUR		6.33	2017	HEAVY	Y	Y	8-20	NORTH & WEST
37	ALONG OMR		NULL	2017	HEAVY	N	NULL	8-20	IT CORRIDOR &
SOUTH									
38	KONDUNAIGAYUR		9.00	2017	HEAVY	Y	N	8-20	IT CORRIDOR &
SOUTH									
39	VYARSARPADI		5.28	2017	HEAVY	Y	Y	8-20	NORTH & LOW L
YING									
40	OTTERI		0.29	2017	HEAVY	N	Y	8-20	NORTH & LOW L
YING									
41	VELACHERY		6.82	2017	HEAVY	Y	O	8-20	CENTRAL & LOW
LYING									
42	T.NAGAR		4.57	2017	HEAVY	N	Y	8-20	CENTRAL & LOW
LYING									
43	GUINDY		7.88	2017	HEAVY	N	N	8-20	ADYAR RIVER B
ANK									
44	SAIDAPET		2.87	2017	HEAVY	N	Y	8-20	ADYAR RIVER B
ANK									
45	MYLAPORE		3.91	2017	HEAVY	N	N	10-12	CENTRAL & LOW
LYING									
46	NUNGAMBAKKAM		3.38	2017	HEAVY	N	Y	18	CENTRAL & LOW
LYING									
47	TAMBARAM		87.40	2017	HEAVY	N	N	8-20	ADYAR RIVER B
ANK									
48	TIRUSULAM		2.40	2017	HEAVY	Y	N	8-20	IT CORRIDOR &
SOUTH									
49	MUDICHUR		3.43	2017	HEAVY	N	N	8-20	ADYAR RIVER B
ANK									
50	ADYAR		6.57	2017	HEAVY	Y	Y	8-20	ADYAR RIVER B
ANK									
51	MADIPAKKAM		5.00	2020	HEAVY	Y	Y	16.3	IT CORRIDOR &
SOUTH									
52	ADAMBAKKAM		3.08	2020	HEAVY	Y	Y	16.3	IT CORRIDOR &
SOUTH									
53	VELACHERY		6.82	2020	HEAVY	Y	O	16.3	CENTRAL & LOW
LYING									
54	TAMBARAM		87.40	2020	HEAVY	Y	N	16.3	ADYAR RIVER B
ANK									
55	MUDICHUR		3.43	2020	HEAVY	Y	N	16.3	ADYAR RIVER B
ANK									
56	GUINDY		7.88	2021	HEAVY	N	N	16	ADYAR RIVER B
ANK									
57	NUNGAMBAKKAM		3.38	2021	HEAVY	Y	Y	21	CENTRAL & LOW
LYING									

58	AMBATTUR	4.50	2021	HEAVY	Y	Y	21	NORTH & WEST
59	OTTERI	0.29	2021	HEAVY	Y	Y	~21	NORTH & LOW L
60	AYANAVARAM	4.61	2021	HEAVY	N	Y	18	NORTH & WEST
61	VILLIVAKAM	3.22	2021	HEAVY	Y	Y	~21	NORTH & WEST
62	EGMORE	2.81	2021	HEAVY	Y	Y	~21+COOVUM BANK	CENTRAL & LOW
63	BABA NAGAR	3.59	2021	HEAVY	Y	Y	~21	NORTH & WEST
64	GKM COLONY	1.00	2021	HEAVY	Y	Y	~21	NORTH & WEST
65	JAWAHAR NAGAR	1.00	2021	HEAVY	Y	0	~21	NORTH & WEST
66	PAPER MILLS ROAD	NULL	2021	HEAVY	Y	Y	~21	NORTH & WEST
67	KUNDRATHUR	9.00	2021	HEAVY	Y	N	~21+HIRISK ZONE	IT CORRIDOR &
68	THIRUMUDIVAKKAM	4.00	2021	HEAVY	Y	N	~21+HIRISK ZONE	IT CORRIDOR &
69	THIRUNEERMALAI	4.80	2021	HEAVY	Y	N	~21+HIRISK ZONE	IT CORRIDOR &
70	SIRUKALATHUR	0.90	2021	HEAVY	Y	N	~21+HIRISK ZONE	NORTH & WEST
71	KORATTUR	7.85	2021	MODERATE	Y	Y	10-17	NORTH & WEST
72	KOLATHUR	11.84	2021	MODERATE	Y	Y	10-17	NORTH & WEST

73	CHETTIMEDU	0.58	2023	HEAVY	Y	N	24	NORTH & WEST
74	RED HILLS	9.92	2023	HEAVY	Y	N	24	NORTH & WEST
75	PERUNGAVUR	3.15	2023	MODERATE	Y	N	24	NORTH & WEST
76	VICHOOR	3.87	2023	MODERATE	Y	N	24	NORTH & WEST
77	TIRUSULAM	2.40	2023	HEAVY	Y	N	24	IT CORRIDOR &
78	INJAMBAKKAM	5.08	2023	HEAVY	Y	0	24	IT CORRIDOR &
79	MUDICHUR	3.43	2023	MODERATE	N	N	24	ADYAR RIVER B
80	T. NAGAR	4.57	2023	LIGHT	N	Y	24	IT CORRIDOR &
81	ALWARPET	4.00	2023	LIGHT	N	Y	24	CENTRAL & LOW
82	EGMORE	2.81	2023	LIGHT	N	Y	24+COOVUM BANK	CENTRAL & LOW
83	ADYAR	6.57	2023	LIGHT	N	Y	24	ADYAR RIVER B
84	THORAIPAKKAM	5.91	2023	LIGHT	N	0	24	IT CORRIDOR &
85	PALLIKARNAI	13.50	2023	HEAVY	Y	N	24	IT CORRIDOR &
86	VELACHERY	6.82	2023	HEAVY	Y	0	24	IT CORRIDOR &

USEFUL TOOLS TO KNOW:

SWD MAPS:

<https://data.opencity.in/dataset/chennai-stormwater-drain-swd-maps>

https://chennaicorporation.gov.in/gcc/swd_net_maps/

ISRO SATELITTE FEED:

<https://bhuvan-app1.nrsc.gov.in/disaster/disaster.php?ld=flood>

References:

2016 floods

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Vettiver Collective; Prema Revathi; Priti Narayan; Students of Madras
Christian College; TN
Labour blog; and V. Geetha.

data opencity gcc swd

2017 floods

the hindu

relief web

2022

Government of India

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2023

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